Intern Experiential Report

Autoliv Japan Ltd.
Autoliv Electronics Japan

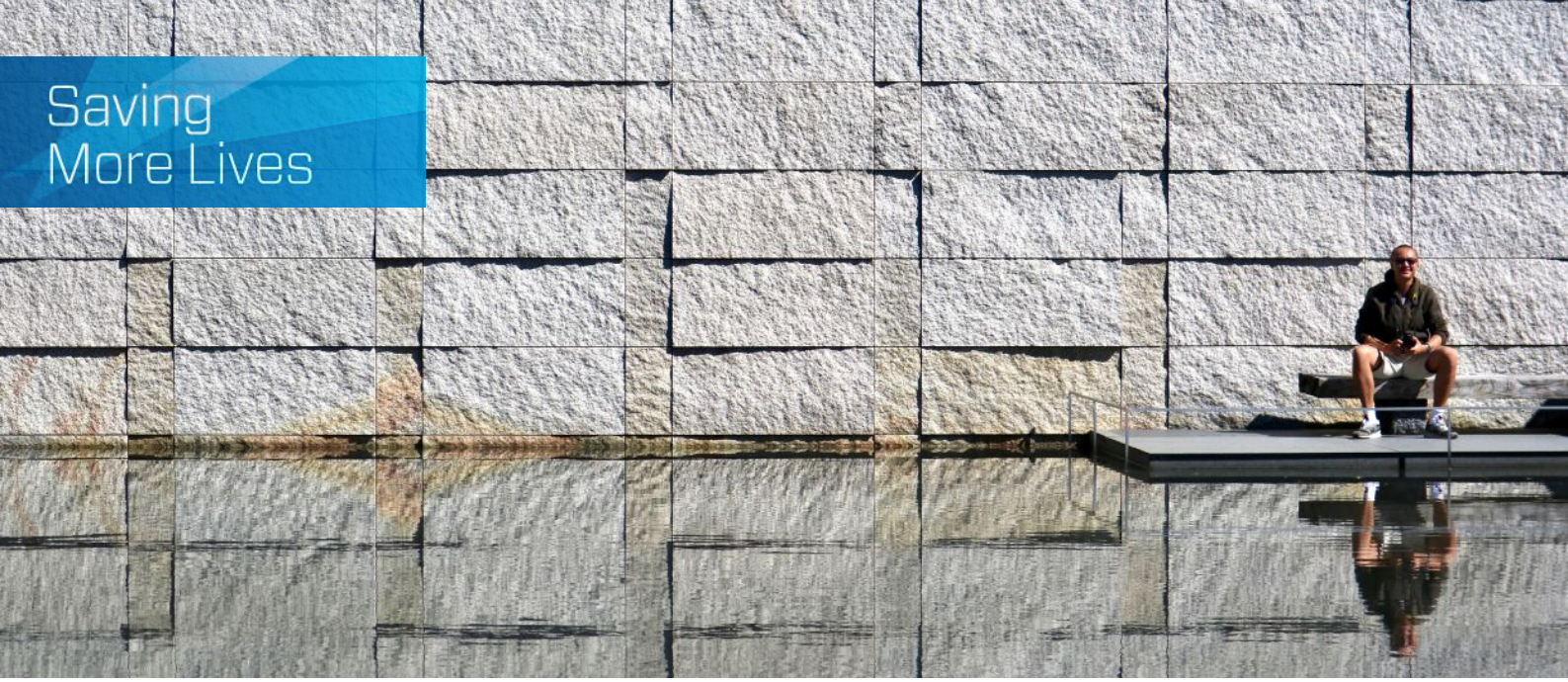
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Roger (Sichen) Luo

The Canada-Japan Co-op Program
The University of British Columbia

December 2016

Autoliv



In Brief

We are at the beginning of the autonomous driving and intelligent transportation evolution, witnessing the growth of advanced sensing, computing, and communication technologies to enable a wide range of applications that change the way we commute, travel, and transport.

A Chinese old saying: "天时，地利，人和" (i.e. Right Time, Right Place, and Right People) perfectly summarizes my 1-year experience as an intern at Autoliv Electronics Japan. Thanks to the Canada-Japan Co-op Program, I am extremely lucky having been worked for a world-class and industry-leading company with a diverse group of talented expertise on several amazing automotive active safety products to share a mission of Saving More Lives.

As a member of the Product Assurance Team, I was able to be involved in a series testing and validation activities of Radar and Vision sensors. These activities have helped me practice the skills I have had as an Electrical Engineering Student, and have motivated me to learn more and to generate new ideas to help the team improve the efficiency. In addition, working with people from other departments on organizing office events have brought everyone fun and excitement.

Woking and living in Japan allows me to get a deeper insight into the automotive industry, transportation systems, advanced technologies, as well as the amazing traditional Japanese culture and craftsmanship. Being exposed myself to these best things have truly inspired me to bring the spirit of Japanese quality in to what I will be doing in the future.

It is an unforgettable and life-changing experience. In this report I will share with you my work experience at Autoliv Yokohama office as well as my overall living experience in Japan. Enjoy!

Roger (Sichen) Luo

Autoliv

Organization

Autoliv is an industry-leading automotive safety systems supplier. Together with its joint ventures, Autoliv has over 80 facilities with 60,000 employees in 29 countries. In addition, the company has 18 development and engineering centres in 9 countries around the world.

Every year, Autoliv's products save over 30,000 lives and prevent ten times as many severe injuries.

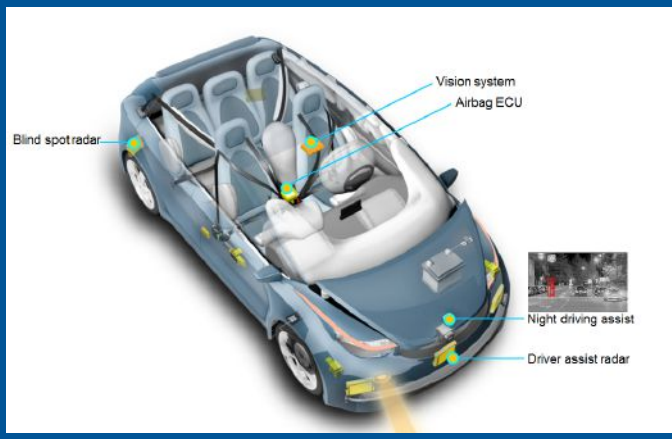


Autoliv Electronics Japan (AEJ) operates 2 close-by offices in Shinyokohma for Active Safety and Passive Safety Engineering. The product include Airbag ECU, Radars, and Vision systems.

These products are developed based on technical requirements from Japanese automobile Original Equipment Manufactures(OEMs) and Japanese traffic conditions and regulations.

The Active Safety Engineering at AEJ consists of teams of Hardware, Software, Radar system, Feature and Function, and Product Assurance.

All of these teams are working together on several Radar sensor and Camera projects for Japanese car manufactures to integrate the product into their vehicles.



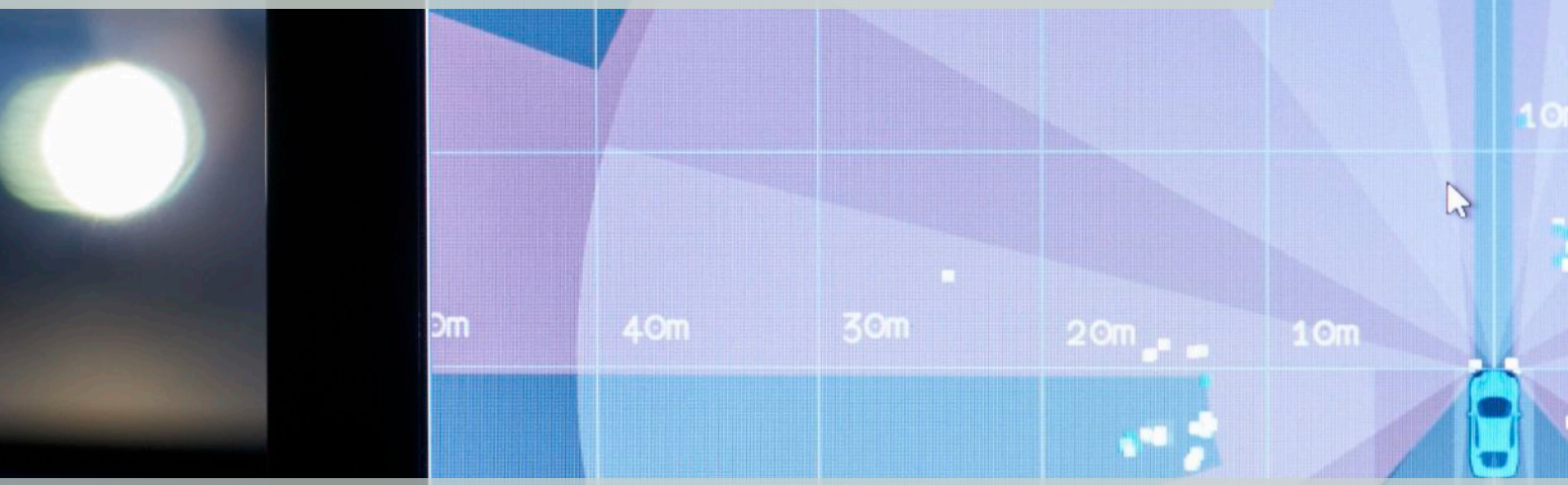


Autoliv Electronics Japan started the partnership with **The Canada-Japan Co-op Program** since 2015.

I am lucky to be one of the first interns from the University of British Columbia together with Chelsea Reid from the University of Victoria to work for the Product Assurance Team from January to December in 2016.

The Product Assurance (PA) team at Autoliv Electronics Japan is responsible for testing and validation of the Radar and Camera products developed for Japanese OEMs.

Test vehicles are prepared with measurement equipment for sensor output visualization and data collection. Collected data are analyzed and evaluated to ensure requirements are met. Software tools are used in the data analysis process.



AS INTERNS, WE NEED TO:

TEST/DEMO VEHICLE SETUP



Equipment Rack
Power Distribution
Measurement Equipment

BENCH TEST



Radar Sensor
Radar Gateway

DATA ANALYSIS



Radar Bumper Effect
Radar Tracking Performance
Radar Cross Traffic Alert
Radar Blind Spot Detection
Vision Lane Detection
Vehicle Dynamics CAN-IMU Output Comparison

DOCUMENTING REPORTING



Data Analysis Criteria
Data Analysis Report
Test Drive checklist
Test Vehicle Specs.
Equipment Prep Manual

TOOL DEVELOPMENT



Wireless ECU Diagnostics and CAN signal Monitoring



A strong foundation under the context of automobile

The exposure to the automotive industry is unique to students from Canadian universities. The Canada-Japan Co-op Program and Autoliv together create valuable opportunities to bring those who are passionate about cars with relative engineering skills to an exciting and futurist world.

Data Management and Analysis

Excel Macros
MATLAB
Python

Computer Software and Programming

Embedded C/C++
Object-oriented Programming
Python
Linux
Real-time



Electronic Hardware

Arduino
Raspberry Pi
Power Distribution
System rack design
Circuit

Other Skills

Team Work
Presentation
Critical Thinking
Technical Report
Email, messaging

Communication Protocol

CAN CAN-FD FlexRay Automotive Ethernet ISO-TP Time-triggered Protocols

Office Events

Valentines Day	Feb. 14
White Day	Mar. 14
Ohanami お花見	Apr. 6
Summer Party	Jul. 9
Winter Party	Dec. 16

White Day



Summer Party

Product Assurance Team Diversity

Canadian	Chinese	Filipinos	French	Greek	Indian	Italian	Japanese	Mexican	Swedish
1	1	1	2	1	1	1	8	1	1

Yokohama
横浜

2nd Largest city in Japan
3M Population
30 min To Tokyo

Intern Living arrangement

The residence is located near the Shinyokohma station, with **18 minute** walking distance to the Autoliv office, and **15 minutes** train ride to Yokohama.

Shinkansen bound for **Nagoya, Kyoto, Osaka** is accessible at the Shinyokohma station.

Nissan Stadium, Nissan Waterpark, Yokohama Arena, Ramen Museum are nearby.





More than an Internship

As a Chinese grown up in Asian culture, Japan seems somehow familiar to me, but at the same time, it is different. Thanks to being able to understand Kanji, in most time I have no problem understanding the signs, labels, and advertisement etc.

Although communicating fluently in Japanese is challenging for me, I am still able to use phrases, gestures and English to communicate with other people.
The friendliness of Japanese people is well-known, this makes my stay in Japan super enjoyable.

The best part of an international internship, is the opportunity of traveling to lots of interesting places.
The opportunity allows us to be loaded with curiosity and an open mind to embrace cultural differences and to understand how their culture shape who they are.

In the following pages I will share with you some of my unforgettable stories.

Never stop exploring















A Life-changing Journey

Kanto

Tokyo
Chiba
Ibaraki
Tochigi
Gunma
Saitama
Kanagawa

Chubu

Gifu
Aichi
Ishikawa
Shizuoka

Kansai

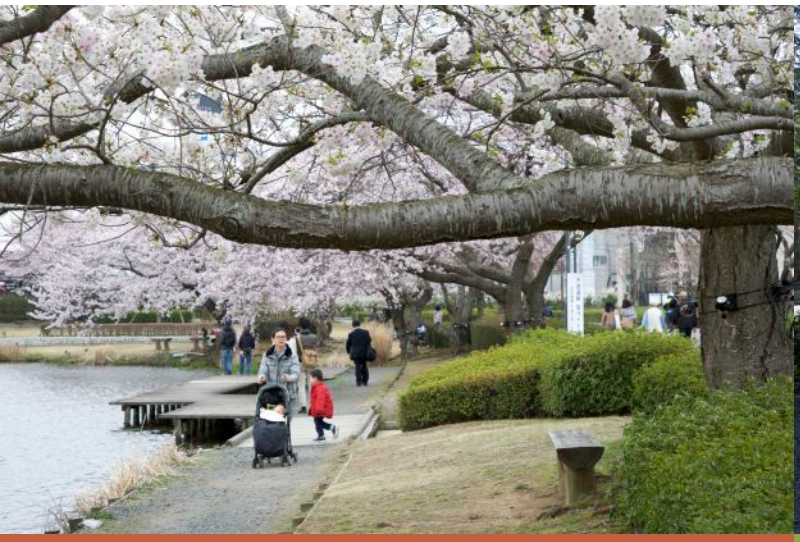
Kyoto
Nara
Osaka

Chugoku

Hiroshima

Hokkaido

Hakodate
Sapporo
Otaru



Cherry Blossom in Mito



Kinomiya Shrine in Atami

Seishun 18 Kippu
青春１８切符

It is a seasonally available ticket offered by JR, giving the rider five days of unlimited, nationwide travel on local and rapid trains for only 11,850 Yen.

It becomes one of my best friends when I was travelling to places that are around 2~3 hours train ride from Yokohama:

Spring:
熱海 Atami,
水戸 Mito,
宇都宮 Utsunomia 日光 Nikko,
甲府 Kofu
高崎 Takasaki,

Summer:
岐阜 Gifu
名古屋 Nagoya
姫路 Himeji



Surprises and joy are found without expectations. With this ticket, I did not make detailed travel plans. Instead, I chose to take random trains going to as farthest as I can, and ended up with the cities on the left.

In most of the time, I could find somewhere that tourists would less likely to go, such as the city Mito. During spring, the Kairakuen and Senba park are filled with cherry blossom and you will see families are having picnic on the grass. In Atami, the Kinomiya Shrine has a tree 大楠 that has over 2000 years history. Nikko is a well-known city by foreigners with lots of shrine and temples. Stopping by Utsunomia to have a set of Gyoza (Dumplings) can't be more enjoyable.

Sometimes the destination would not have too many tourist places, but taking time to enjoy the train ride can really settle down the busy mind to be in the moment.







Fuji -Q HIGHLAND & DISNEY SEA

These 2 amusement parks are truly unique to Japan because of the view of Mt. Fuji and the immersive experience of the myths and legends of the sea.

There are four roller coasters at Fuji-Q Highland and each features a world record spec such as height, length, speed, or steepness.

The Disney Sea is made of seven themed ports. Good selections of food are available at each each port. Fast pass is available for popular rides at Disney Sea, and Single Rider Line can reduce the time in line from 2 hours to 15 minutes.







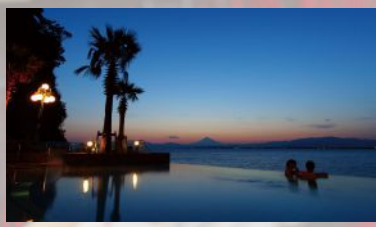


Kamakura 鎌倉

A coastal town in Kanagawa Prefecture, about 30 minutes train ride from Yokohama and less than an hour from Tokyo.

It is sometimes called the Kyoto of Eastern Japan, offering numerous temples, shrines and many other historical monuments.

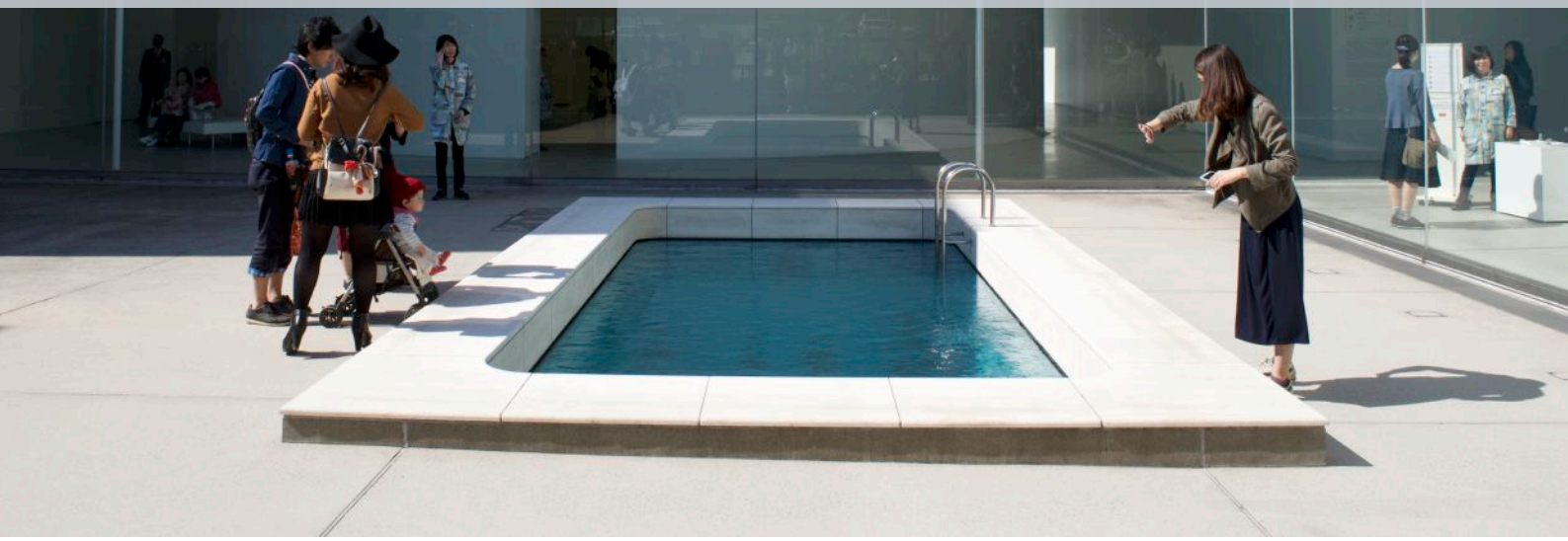
On the island Enoshima 江ノ島 near Kamakura, a hot spring spa has a infinity pool offering a nice view of the ocean.

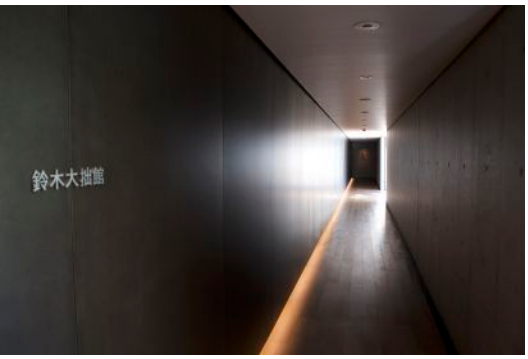


Kanazawa 金沢

The city features a number of modern museums such as the 21st century museum shown in the background. Created with a piece of glass and a thin layer of water on the top, people appear to be underwater when looking down from outside from the “swimming pool”.

The main attraction Kenrokuen 兼六園 is consider as one of the three best landscape gardens in Japan.














愛媛県川之江市上分町 加藤絹子
寝屋川市点野二丁目十六ノ一 川上詩朱印
渋谷区 (株)ワールドイン 吉田智
田川市糒一六九二ノ七 羽音一枝
名古屋市 株式会社あいち研醸社
大阪市淀川区 株式会社V・Wテック 代表取締役 筒井孝一
城陽市 人見知賀子 森田康雄 洋子 珠須
伊丹市北河原五丁目三番十二号
栃木県佐野市金屋仲町二、四三〇 安蘇土木建築(株) 落合勝敏
名古屋市北区元志賀町二丁目一七ノ一 貝塚市 岩﨑清彦
東京都世田谷区 株式会社グラディー 佐沢一馬
東京都足立区 (有)沙羅詩屋 古性秋光
大阪府共和町三ノ二ノ一〇 有中村広報 中村直也
サンピエール病院 山崎學
交野市 (株)和建設 山田登子 寛台邨
東京都中央区築地 魚がし (株)浦安田代水産 齋藤進
京都府南区九条上殿田町 (有)ワンダーランド 野口大輔
宮崎県日南市 医療法人 同仁会 谷口病院 谷口浩
名古屋市千種区 (株)川島鉄工所 小林和代
北海道石狩郡当別町園生五五番地二九 (株)シャローム アン 上原貴

Kansai
関西



As a cultural and historical heart of Japan, Kansai is famous for the beauty of the tourist attractions in Kyoto, the food in Osaka, the historic treasures in Nara, and so much more. During the golden week, I spent 3 days in the Kansai area, trying to find the difference from the Kanto area I was living in.

Kyoto

Walking on the streets of Kyoto feels completely different from walking on busy shopping areas like Ginza, Shibuya and Shinjuku in Tokyo.

Temples, shrines, parks are everywhere and if you missed one, there will be another one within a minute of walk.

Because time is limited, I only stayed in Kyoto for 1.5 days. But I have still visited a few popular places such as
Fushimi Inari-taisha
伏見稲荷大社

Kiyomizu-dera
清水寺

Kinkagu-ji
金阁寺

Sanjosangen-do
三十三間堂

Nara

A one-day bus pass is available at the JR Nara station.

The bus covers all of the historical attractions including the most famous Todaiji Temple東大寺,

Horyuji Temple 法隆寺, Nara

Park, Kofukuji Temple, and so much more.

The weather turns rainy in the afternoon on the day I traveled to Nara.
When I stopped by a small restaurant eating udon, the owner of the restaurant, an old lady, offered me an umbrella and let me to take it back home. This is one of the many moments that I can feel the care from Japanese people.

Osaka

The first things everyone will notice is people are standing on the right side of the escalator instead of the left side in Tokyo. Here is what I found online:

During the Tokugawa Period, Edo (Tokyo) was a city of samurai, who preferred to be on the left so they could draw their swords easily. Osaka, on the other hand, was a city of rich merchants, who preferred to be on the right so they can hold their abacus.

When getting off the train at the Osaka station, the Umeda shopping mall 梅田地下街 is underground. I

was impressed by how enormous it is and got lost when transferring the train.

Kyoto








Nara






Osaka











Mt. Fuji
富士山

The students from the Canada-Japan Co-op Program teamed up for a hike to Mt.Fuji when it was opened for public hikers.

We took buses at noon from Tokyo/ Yokohama, arriving at the 5th station - the starting point of hiking in the afternoon.

The hike starts at 19:30 when the sunset start. With headlight, snacks, jacket, gloves, and support from everyone in the team, we arrived at the summit just when the sunrises on the next day.

This experience made me mindfully appreciate the beauty of nature.

4:30 Jul. 11

During the hike, there are huts at each station for hikers to rest or stay overnight. We stopped by each of the station for short breaks.

As we go higher, the temperature became lower and people were getting tired, but this did not stop us.

19:26 July 10



Hikers

3:00



天时 地利 人和

Right Time Right Place Right People

The people I have met throughout the year give me positive influence in many different ways.

The company consists of employees from all over the world with different ages, color, religion, and culture background. Talking to them brought me fresh ideas about the way we look at things, and also made me more open-minded to embrace the difference of each individual. The understanding of inclusion and diversity is extremely important especially when developing products that serve for every walk of life.

The Canada-Japan Co-op Program made it possible for me to meet a group of fantastic students from all places in Canada, making our stay in Japan full of joy.

When travelling, there are countless Japanese people who offered me help on finding the ways and giving recommendations of tourists places and restaurants. Their care and attention to details are fully embedded to everything they do and every product they make. These people become the role models that I should keep studying from.

Finally, meeting my family and friends coming from my hometown allowed me share my excitement to them.

Since I moved to Canada in 2013, I consider Vancouver as my second home after my hometown Kunming. Now there is no doubt that Japan have become my third home.

Tokyo Skytree

















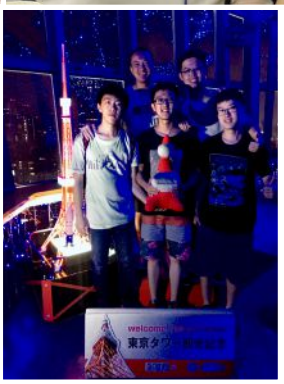




















For the love of transportations

Humans build tools.

In a small way, the software tools we build at company can increase the efficiency of processing data. In a big way, the cars, trains, airplanes, ferries, and even the elevators we use daily are making our lives easier.

There are so many types of transportations moving people of different races and ages around. These tools enable humans to commute, travel, doing business, meeting friends and family. Before the Chinese new year each spring, people return home from work or studies to have Reunion dinner with their families on New Year's Eve.
This migration creates a 2.9 billion (2016) of load on transportation.

The airplane makes it possible for me to pursue my undergraduate studies in Canada, and this year it took me to Japan. Throughout the year, the well-integrated, advanced public transportation system in Japan allowed me to travel to the places you have just seen. I also had chances to visit a number of train museums, car museums, and airplane maintenance facilities to get a deeper insight into how these things work.

The exposure to these transportation solutions make me deeply appreciate humans' intelligence of making tools to do great things. The products we develop at Autoliv will be installed on hundreds and thousands of cars to bring safety features that can potentially save lives, which is very motivating. I feel proud of being a part of the team.



We are at the beginning

In another 10 years

Before the iPhone was introduced 10 years ago, I never thought the evolution of smartphone can be this fast that majority of the cellphones shipped today are running advanced operating systems. The rise of smartphone also causes a ripple effect, creating opportunities for app developers, network carriers, and other countless service providers to take advantage of the mobile platforms.

The one-year internship have convinced me that a similar revolution is now happening in the automotive industry, or put it larger, in the transportation domain. With more advanced sensors, more powerful computer chips and software, and faster communication technology available, vehicles are becoming much smarter than ever before. We are only at the beginning of this change.

I feel lucky having had a sneak peak into the industry and now I become even more passionate about what I can do. I mark this experience in Japan as a milestone of my lifetime, because it has defined an important new direction for my future career.

In the next 10 years, there will be a lot more changes and unknowns that I need to deal with, but I will stay motivated and humble to learn from the people around me, and share my experience to help other people.

ありがとうございました！



10 Years
2017 - 2027

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Thank you!

Photo Credit:
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Chelsea Reid
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(Dept. of Electrical and Computer Engineering)



THE CANADA-JAPAN CO-OP PROGRAM

THE UNIVERSITY OF BRITISH COLUMBIA

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Products save over
30,000 lives
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